

Machine Learning Classification of Supernova Neutrinos in Noisy Detector Environments

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ABSTRACT: This work investigates the classification of simulated supernova neutrino interactions in liquid argon detectors using machine learning. Detector performance is limited by electronic noise, which obscures signal features. Logistic regression and dense neural networks achieve near-perfect accuracy in the absence of noise, but degrade rapidly as noise increases. A convolutional neural network performs better at low noise, reaching an accuracy of 0.962 at $\sigma = 5$. It then collapses to random guessing at higher noise levels. Alternative architectures such as Inception CNNs provide only limited improvement. In contrast, localised Gaussian blob backgrounds have minimal impact, with accuracies remaining above 0.98 across most parameter ranges. These results show that classification performance is primarily limited by the signal-to-noise ratio rather than model complexity.

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1 Introduction

1.1 Supernova neutrinos and scientific motivation

Supernovas are violent stellar explosions that mark the end of the life cycle of massive stars. In addition to emitting intense electromagnetic radiation, they produce a large burst of neutrinos. A well-known example is SN1987A [1], a type II supernova observed in 1987, which produced a detectable signal several hours before the arrival of visible light. This occurs because neutrinos interact only weakly with matter and can escape the dense stellar core more easily than photons.

These neutrinos carry information about the internal dynamics of the collapsing star and therefore provide a unique probe of supernova physics. Furthermore, their early arrival makes them particularly valuable for early-warning systems. The SuperNova Early Warning System (SNEWS) aims to detect such neutrino bursts in real time, enabling rapid follow-up observations across multiple astronomical channels. Reliable identification of supernova neutrino signals is therefore of significant importance in both particle physics and astrophysics.

1.2 Neutrino interactions

Neutrinos are electrically neutral leptons with extremely small masses. They exist in three flavours: electron (ν_e), muon (ν_μ), and tau (ν_τ), each associated with a corresponding charged lepton (e^- , μ^- , τ^-). Due to their lack of electric charge and their weak interaction cross-section, neutrinos interact only via the weak nuclear force. This low interaction probability makes them difficult to detect.

Neutrino interactions can be categorised into charged-current (CC) and neutral-current (NC) processes. In charged-current interactions, the neutrino is converted into its corresponding charged lepton through the exchange of a $W^\pm$ boson. For example,

$$\nu_e + n \rightarrow e^- + p, \tag{1.1}$$

where a neutron is converted into a proton and an outgoing electron is produced. The charged lepton typically leaves a visible track in the detector, making CC interactions easier to identify.

In contrast, neutral-current interactions are mediated by the Z^0 boson and do not change the neutrino flavour. An example process is

$$\nu + N \rightarrow \nu + N, \tag{1.2}$$

where the neutrino scatters off a nucleon N . These interactions do not produce a charged lepton, and detection relies on secondary signals such as nuclear recoil, making them more challenging to observe.

1.3 Liquid argon time projection chambers

Modern neutrino experiments, such as MicroBooNE [2] and the Deep Underground Neutrino Experiment (DUNE) [3], employ liquid argon time projection chambers (LArTPCs) as their primary detection technology. In these detectors, a large volume of liquid argon is subjected to a uniform electric field [4]. When a neutrino interacts within the detector, it produces charged particles that ionise the argon atoms along their paths. A dominant interaction for electron neutrinos in argon is the charged-current absorption process,

$$\nu_e + {}^{40}_{18}\text{Ar} \rightarrow e^- + {}^{40}_{19}\text{K}$$

where the incoming neutrino converts into an electron and excites the argon nucleus. The outgoing electron produces a visible ionisation track, while the excited nucleus may subsequently de-excite via photon emission.

The freed electrons drift under an electric field and the signals produced by these drifts are collected on a detection plane. By measuring the arrival time and position of these electrons, a high-resolution three-dimensional image of the particle trajectories can be reconstructed. In addition to ionisation charge, scintillation light is also produced promptly, providing timing information that helps localise the event within the detector volume.

Both the ionisation and the scintillation signals enable precise reconstruction of the interaction including the particle tracks and energies. However, for supernova neutrinos, which typically have energies in the few MeV range, the resulting signals are relatively faint. This leads to sparse images with low pixel intensities, making them harder to distinguish from background noise.

An example of a simulated LArTPC image is shown in Figure. 1. The visible structure corresponds to the ionisation track of a charged particle produced in a neutrino interaction, with pixel intensity indicating the amount of deposited energy.

2 Dataset and Physics Context

2.1 Simulated image dataset

The dataset used in this study consists of simulated liquid argon time projection chamber (LArTPC) images representing supernova neutrino interactions. Each event is stored as a two-dimensional image, where the pixel intensity corresponds to the ionisation charge deposited in the detector.

Supernova neutrinos typically have energies in the few MeV range, resulting in relatively faint and sparse detector signals. As a result, the corresponding images contain only a small number of activated pixels, often forming short and localised track-like structures. This low signal strength presents a significant challenge when attempting to distinguish genuine neutrino interactions from detector noise.

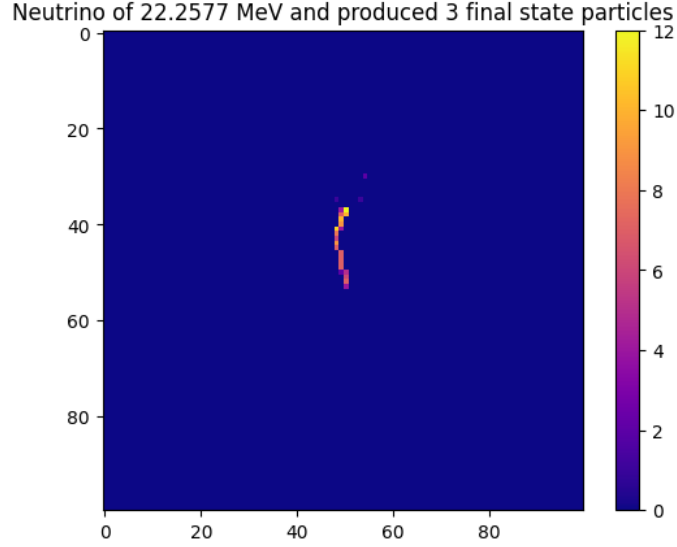


Figure 1. Example simulated LArTPC image of a neutrino interaction. The colour scale represents deposited energy, with brighter pixels corresponding to higher ionisation charge. The sparse structure reflects the low-energy nature of supernova neutrino interactions.

2.2 Metadata and target quantities

Each image is accompanied by metadata describing the particles involved in the interaction, including particle identification (PDG codes [5]), energies, and momentum components. For the primary classification task, only the image data are used, and the goal is to distinguish between neutrino interaction events (signal) and artificially generated noise events (background).

The metadata are utilised in later extensions of this work, where quantities such as neutrino energy and electron energy are extracted and used as targets for regression models. This enables an investigation of energy reconstruction and resolution.

2.3 Initial data exploration

To characterise the dataset, the distribution of pixel intensities was analysed. The maximum pixel value of each image was used as a simple measure of total energy deposition, since higher intensities correspond to larger ionisation signals.

Figure 2 shows that most neutrino events have relatively low maximum pixel values, with a narrow distribution concentrated at small intensities. This confirms that the signal is intrinsically weak and motivates careful consideration of noise levels when constructing background datasets.

2.4 Simulation of electronic noise

The Liquid-argon time projection chambers (LArTPCs) are sensitive to electronic noise arising from the detector itself. This noise is present across the entire detector and therefore is present uniformly across the recorded images. Accurately modeling this noise is

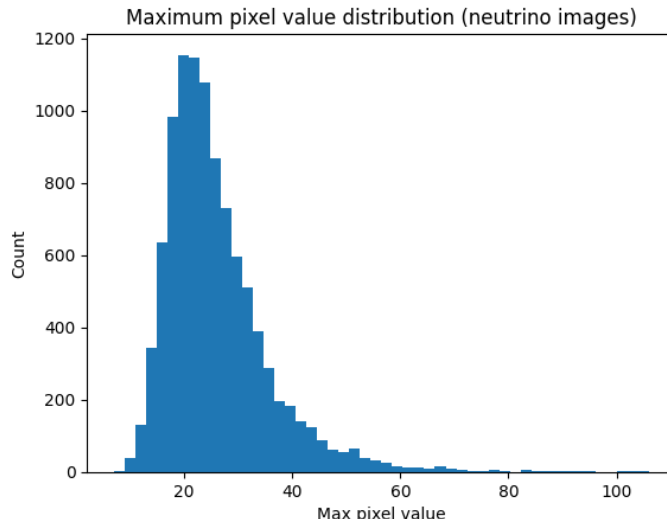


Figure 2. Distribution of maximum pixel values for neutrino interaction images. The majority of events exhibit low pixel intensities, reflecting the low-energy nature of supernova neutrino interactions.

essential to testing our classification algorithms, as it determines how efficiently each classifier distinguishes true neutrino signals from background.

In this project, electronic noise is simulated by sampling from a Gaussian (normal) distribution with the mean centred at zero. The noise contribution, n , is therefore given by

$$n \sim \mathcal{N}(0, \sigma^2)$$

where σ controls the strength of the noise. The noise image is constructed as

$$I_{\text{noisy}}(x, y) = I_{\text{signal}}(x, y) + n(x, y)$$

followed by clipping to ensure physically meaningful pixel values (i.e. non-negative intensities)

A zero-mean Gaussian distribution is used since electronic noise arises from many small, independent fluctuations. A mean of zero ensures no systematic bias in pixel intensity therefore introducing fluctuations about the true pixel intensity [6].

By varying, σ , the standard deviation of the distribution it is possible to simulate different noise regimes. Regimes such as low-noise environments where signal dominates to high-noise environments where fluctuations become comparable to true neutrino signal intensities. This behaviour is illustrated in Figure 3 which shows that as the noise level increases, the maximum pixel values of noise-only images increase. At sufficiently high noise levels, the distribution begins to overlap with that of the neutrino signal.

Neutrino interactions typically produce localised high-intensity pixel clusters, whereas noise is more uniformly distributed. Increasing σ raises the likelihood that noise mimics signal-like features, thereby challenging the classifier and enabling investigation of failure modes such as false positives and reduced detection efficiency.

The Gaussian approximation captures the dominate random behaviour of electronic noise and is efficient to compute making it an appropriate starting point for this analysis[7].

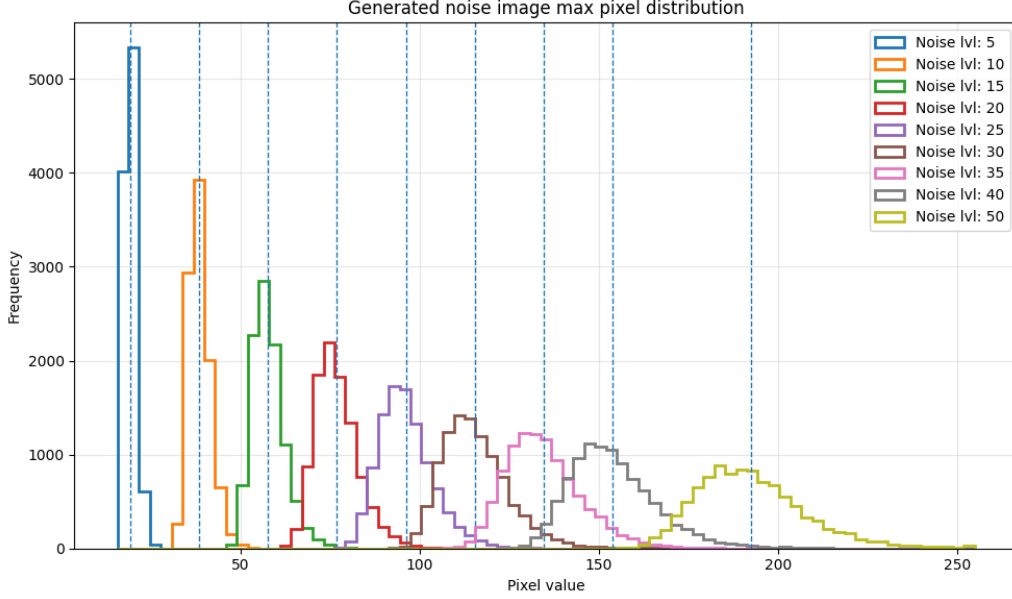


Figure 3. Distribution of maximum pixel values for noise-only images at different noise levels. Increasing the noise level broadens the distribution and shifts it towards higher intensities.

2.5 Construction of classification datasets

Classification datasets were constructed by combining signal and background images into two classes:

- **Signal:** neutrino interaction images, with or without additional noise.
- **Background:** images consisting purely of simulated noise.

In addition to noise-only images, datasets were generated in which electronic noise was overlaid directly onto neutrino images. This provides a more realistic representation of detector conditions, where genuine signals are contaminated by noise.

All images were normalised prior to training to ensure numerical stability during optimisation while preserving the relative structure of the signal.

3 Machine Learning Methodology

3.1 Binary classification task

As mentioned the primary objective of this study is to distinguish between simulated neutrino interactions and background noise using supervised learning techniques.

Each image is assigned a binary label, where true neutrino events are treated as signals (class 1) and noise-only images as background (class 0).

This defines a standard binary classification problem, where the goal of the model is to learn a mapping from pixel intensities to class labels. In this section only image data are being used as input, ensuring that the classification is solely based on spatial and intensity features.

The challenge of this task, as seen in [7], arises from the low-energy neutrinos, which produce faint signals comparable in scale to electronic noise. As a result the classification limit is strongly dependant on the noise levels.

3.2 Baseline models

To establish a performance baseline, two simple models were implemented prior to more complex architectures.

Logistic regression provides a linear classifier that models the probability of an image belonging to the signal class via a sigmoid activation function[9],

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad (3.1)$$

where $z = \mathbf{w} \cdot \mathbf{x} + b$. Here, $\mathbf{x}$ represents the input features (pixel values of the image), $\mathbf{w}$ is a vector of learnable weights that determine the importance of each feature, and b is a bias term that shifts the decision boundary. While being easy to compute, it assumes a linear decision boundary and therefore cannot capture complex spatial features within the image. This process is illustrated schematically in Figure. 4.

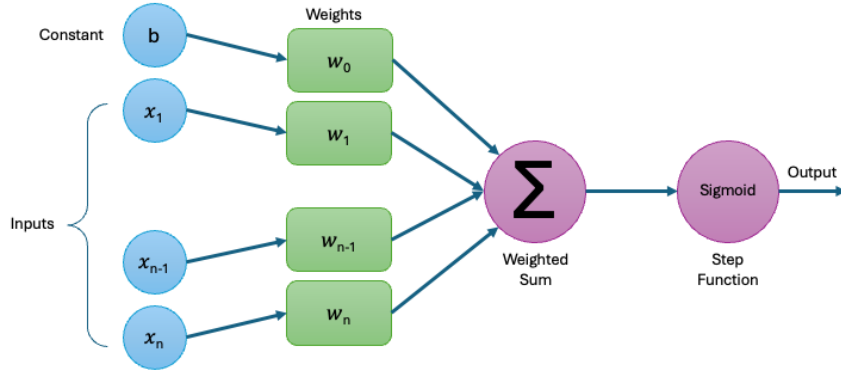


Figure 4. Schematic representation of a single neuron used in logistic regression and dense neural networks. Input features x_i are multiplied by weights w_i , summed with a bias term b , and passed through a sigmoid activation function to produce an output probability.

Dense neural network extends this approach by introducing multiple layers of interconnected neurons, where each neuron computes

$$y = f\left(\sum_i w_i x_i + b\right), \quad (3.2)$$

with f a non-linear activation function [10]. This enables the model to learn more complex decision boundaries, although spatial information in the image is not explicitly preserved.

3.3 Convolutional neural network architecture

Convolutional neural networks (CNNs) work well for this task because they make use of the spatial structure in images [8].

Each convolutional filter is a small matrix of learnable parameters that is applied across the input image to produce a feature map. During training, these filters adapt to respond to specific spatial patterns in the data. In early layers, filters typically identify simple structures. For neutrino images, the filters respond strongly to localised high-intensity clusters. In deeper layers, combinations of these features allow the network to recognise more complex structures. Structures such as short track-like patterns characteristic of neutrino interactions. Pooling layers then reduce the size of the data, making the model more efficient to run. The behaviour of convolutional filters can be understood by looking at their response to signal and noise images, as shown in Figure. 5.

For noise-only images, the responses are more spread out and lack structure. This reflects the random nature of noise. It also shows how the CNN separates signal from background using spatial patterns.

The CNN used in this work is built from convolutional and pooling layers, followed by fully connected layers for classification. The model includes:

- Convolutional layers with small 3×3 kernels to detect local features
- More filters in deeper layers to capture more complex patterns
- Max-pooling layers to reduce image size
- Fully connected layers to make the final prediction
- A sigmoid output neuron for binary classification

and is shown in Figure. 6.

ReLU activation functions are used in the hidden layers to introduce non-linearity. The model is trained using binary cross-entropy loss and an adaptive optimiser. This structure helps the network learn step by step, starting with simple pixel features and building up to more complex patterns linked to neutrino interactions. Similar approaches are widely used in high-energy physics, where spatial patterns are important for distinguishing signal from background [7].

3.4 Performance metrics

Model performance is assessed using a range of different metrics. Accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3.3)$$

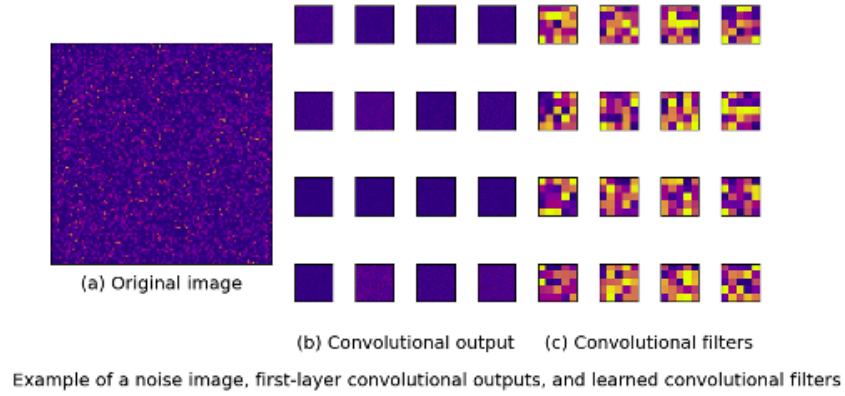
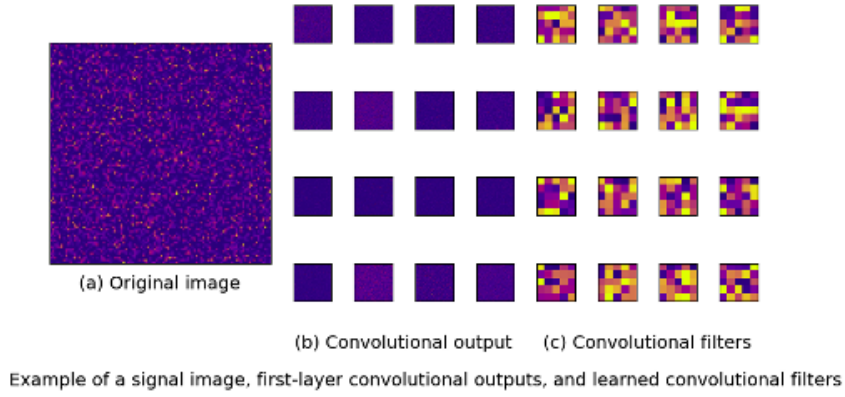


Figure 5. Example of a neutrino interaction image (left), corresponding first-layer convolutional feature maps (centre), and learned convolutional filters (right). The filters respond to localised high-intensity structures characteristic of neutrino events.

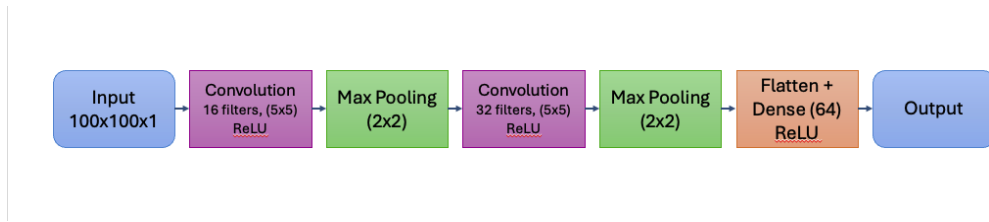


Figure 6. Schematic of the convolutional neural network architecture used in this work. The model consists of two convolutional layers with ReLU activation, each followed by max-pooling, before flattening and fully connected layers leading to a sigmoid output for binary classification.

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives respectively. The false positive rate (FPR) and false negative rate (FNR) are

defined as

$$\text{FPR} = \frac{FP}{FP + TN}, \quad \text{FNR} = \frac{FN}{FN + TP}. \quad (3.4)$$

These metrics have direct physical interpretations. A high FPR corresponds to noise being misidentified as neutrino events (reduced purity), while a high FNR corresponds to genuine neutrino interactions being missed (reduced efficiency). Confusion matrices are used to visualise classifier performance and to identify systematic biases in predictions [7].

4 Electronic Noise Classification Results

4.1 Visualising electronic noise

Example images at increasing noise levels are shown in Figure. 7. As the noise parameter, σ increases, fluctuations appear across the full detector plane. At high noise levels, signal features become comparable in scale to these fluctuations which reduces the contrast between signal and background. This makes classification more challenging and reduces accuracy.

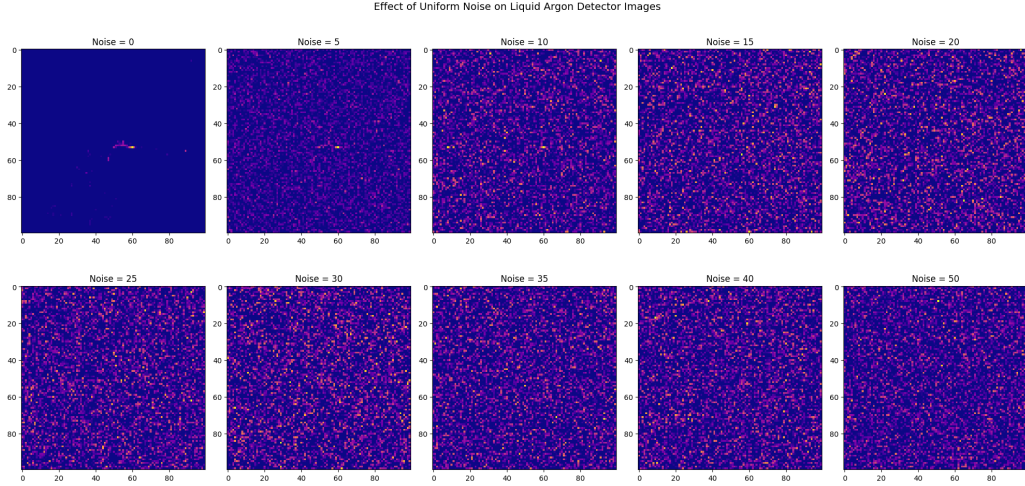


Figure 7. Images showing the effects of increasing noise levels across the full detector plane on a genuine neutrino event

4.2 Pixel-value and maximum-pixel distributions

The classifiers' behaviour can be understood by comparing the maximum pixel value distributions of the signal and the noise as illustrated in Figure. 8. As shown in section 2, neutrino events occupy a narrow range of low intensities, while increasing noise broadens and shifts the background distributions. At higher σ , the background distribution broadens and shifts towards higher pixel values. At intermediate values of σ , the noise distribution overlaps significantly with the signal distribution. This increases the likelihood of misclassification. Finally at very high values of σ the background becomes dominated by large fluctuations and the physically signal gets effectively obscured.

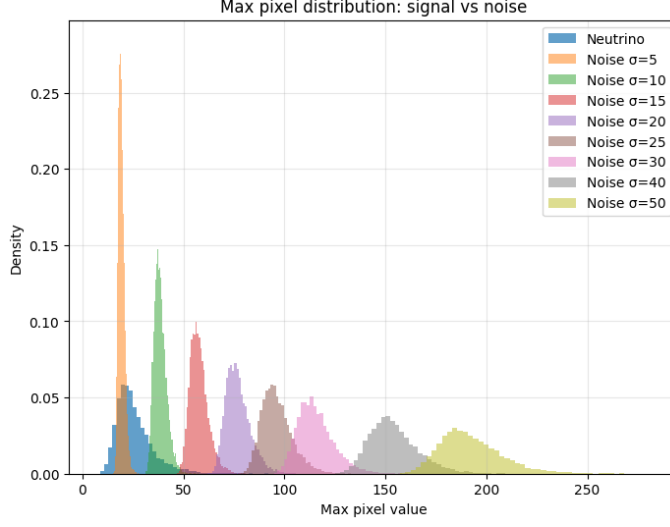


Figure 8. Maximum pixel value distributions for neutrino (signal) and noise-only images at different noise levels σ . Increasing σ broadens and shifts the noise distribution towards higher pixel values. At intermediate noise levels, significant overlap with the signal distribution is observed.

4.3 Baseline model performance

The first models tested were logistic regression and a dense neural network. These provide useful baselines because they do not explicitly preserve spatial structure in the input image. For clean images, both baseline models perform very well. Logistic regression achieved an accuracy of 0.999 at zero added noise, while the dense neural network achieved 1.000. This shows that, without noise in the system both classifiers can easily distinguish between signal and noise events.

As the noise level increases, however, the performance of both baseline models degrades steadily. Logistic regression falls from 0.999 at $\sigma = 0$ to 0.863 at $\sigma = 5$, 0.664 at $\sigma = 10$, and approaches random guessing at the highest noise levels, with accuracies near 0.5. The dense neural network follows a similar trend and performs slightly worse over much of the scanned range, dropping to 0.610 by $\sigma = 10$ and then also approaching 0.5 at high noise. Both models therefore become increasingly sensitive to global pixel fluctuations as the background grows stronger.

4.4 CNN performance across noise levels

The convolutional neural network performs best at low noise, where spatial information remains informative. It achieved an accuracy of 1.000 at zero noise and 0.962 at $\sigma = 5$, outperforming both baseline models in this regime.

The CNN results also show that performance does not decline smoothly with increasing σ . Beyond $\sigma = 10$, the measured accuracy falls to 0.500, indicating that the network has effectively collapsed and is performing random guessing.

The results in sections 4.3 and 4.4 can be clearly shown in Figure 9.

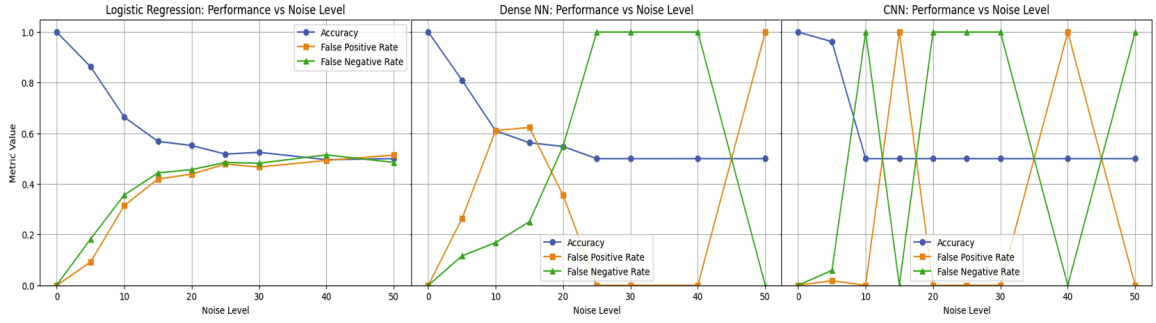


Figure 9. Performance of the logistic regression, dense neural network, and baseline convolutional neural network as a function of noise level σ . Each panel shows the accuracy, false positive rate, and false negative rate. The logistic regression and dense neural network degrade steadily as the noise increases. The CNN performs best at low noise, but becomes unstable at higher noise levels, where the false positive and false negative rates can become extreme.

4.5 Interpretation in the presence of detector noise

In this project, both signal and background classes include noise at the same level, reflecting the fact that detector noise affects all recorded events.

The results presented in Sections 4.3 and 4.4 show that classifier performance depends strongly on the noise level. At low values of σ , the structure of neutrino interactions remain clearly distinguishable.

As the noise level increases, the amplitude of random fluctuations becomes comparable to the signal itself. This reduces the contrast between the neutrino interactions and the surrounding detector plane. At sufficiently high noise levels, both signal and background images are dominated by random fluctuations. In this regime, the classifier can no longer reliably identify meaningful structure. This leads the classifiers to instead exhibit unstable behaviour. This includes large false positive or false negative rates and, in the case of the CNN, a tendency to guess randomly. This difference arises from the way each model processes the image data. Models that do not explicitly preserve spatial structure rely primarily on intensity information. This makes them more sensitive to uniform fluctuations in pixel values. In contrast, the CNN is able to exploit local spatial correlations, allowing it to identify the clustered, track-like features characteristic of neutrino interactions.

5 Extensions

5.1 Alternative CNN architectures

To test whether the baseline CNN could be improved, two alternative architectures were studied. These were an Inception-style CNN and a residual CNN. Both are widely used in image classification and address known limitations of standard CNNs. Inception modules allow the network to analyse features at different spatial scales. Residual connections help information pass through deeper networks and improve training stability [12, 13].

The Inception CNN utilises an Inception module, as shown in Figure. 10. This module applies multiple convolutional filters in parallel, allowing the model to capture both small local features and larger-scale patterns. This improves feature extraction by enabling multi-scale analysis within a single layer.

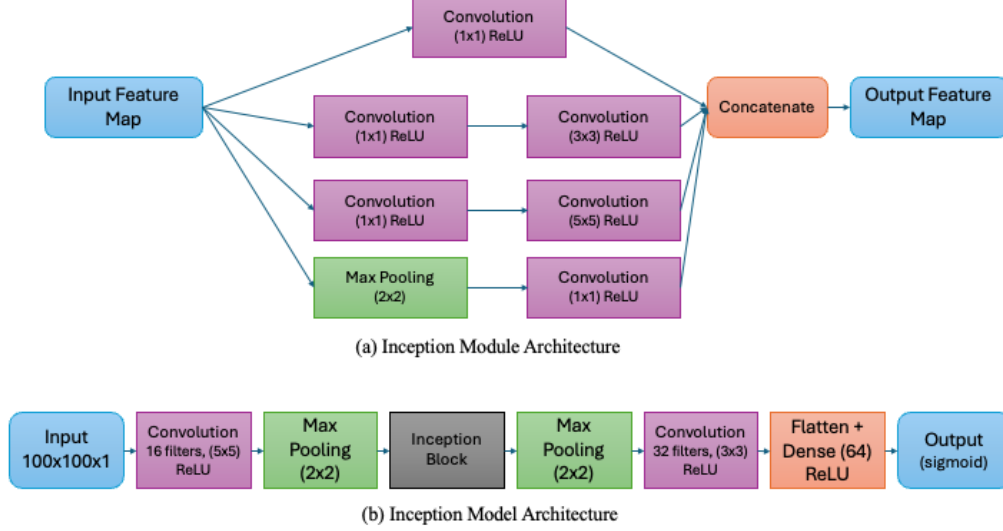


Figure 10. (a) Inception module showing parallel convolutional paths with different kernel sizes. (b) Full Inception CNN architecture used in this work.

The residual CNN utilises a residual block that adds the input feature map to the output of two convolutional layers, as shown in Figure. 11. This enables the network to learn residual corrections rather than a full mapping. This improves training stability and performance in deeper models [13].

The performance of all three CNN models is shown in Figure. 12. At low noise, all models perform very well. Each model achieved an accuracy of 1.000 at $\sigma = 0$. The Inception and residual CNNs both retained an accuracy of 0.962 at $\sigma = 5$. Differences appear at intermediate noise levels. At $\sigma = 10$, the Inception CNN achieved an accuracy of 0.751, while the residual CNN dropped to 0.500. This suggests that multi-scale features improve robustness

At higher noise levels, all architectures approach random guessing, with accuracies tending towards ~ 0.5 and extreme false positive or false negative rates. This behaviour is similar to the baseline CNN results in Section 4.

These results show that architecture changes do not remove the main limitation. The Inception CNN delays this breakdown slightly, which is expected since neutrino events contain structure. However, the overall performance is still limited by the signal-to-noise ratio rather than the model design [7].

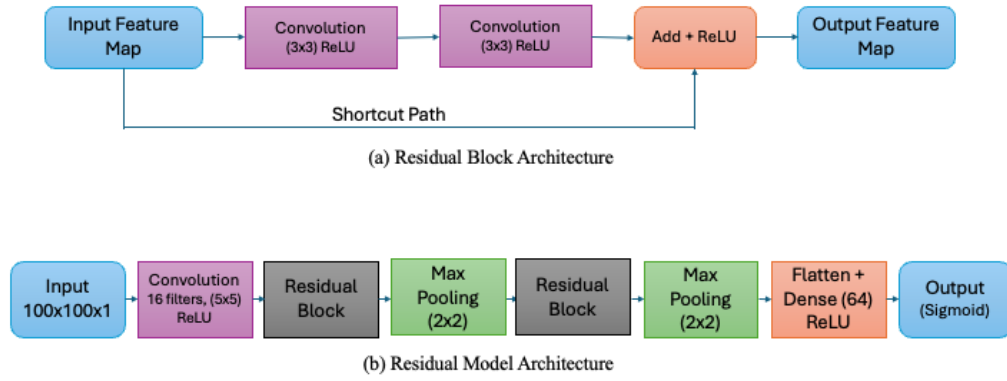


Figure 11. (a) Residual block with shortcut connection. (b) Full residual CNN architecture used in this study.

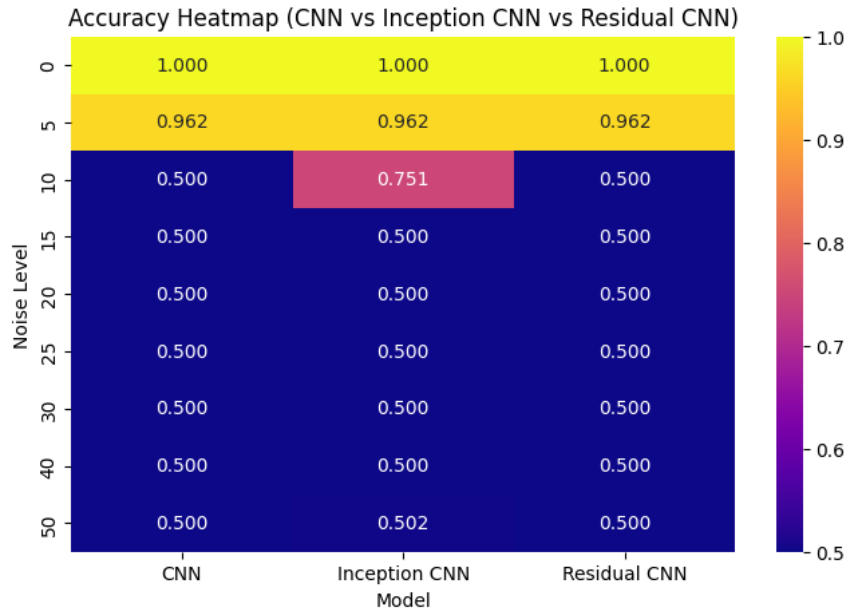


Figure 12. Classification accuracy as a function of noise level for the baseline CNN, Inception CNN, and residual CNN. All models perform well at low noise, but accuracy degrades as noise increases, approaching random guessing.

5.2 Radioactive noise simulation with Gaussian blobs

5.2.1 Gaussian blob simulation

In addition to electronic noise, detectors can be affected by radioactive backgrounds. Radioactive backgrounds were modelled as localised Gaussian features, motivated by low-energy β emitters such as ^{39}Cl , ^{39}Ar , and ^{35}S . These produce small, localised energy depositions that may resemble weak neutrino signals.

Each event is described by a two-dimensional Gaussian,

$$I_{\text{blob}}(x, y) = A \exp \left[-\frac{(x - x_0)^2 + (y - y_0)^2}{2\sigma^2} \right], \quad (5.1)$$

where (x_0, y_0) is the centre, A is the peak intensity, and σ controls the width. The centre is chosen randomly within the image, while A and σ are varied.

Examples of simulated blob images are shown in Figure. 13.

Radioactive Gaussian blob images
sigma = 3, height = 5

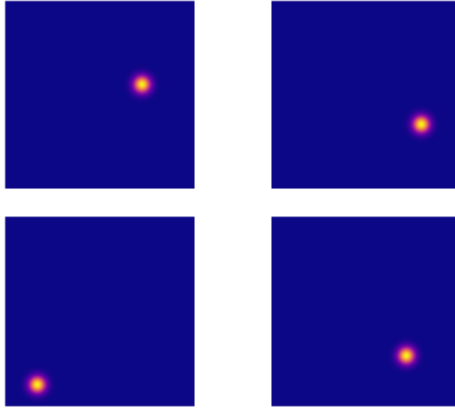


Figure 13. Examples of simulated Gaussian blob images used to model radioactive backgrounds. The blobs are localised in space and vary in position and intensity.

5.2.2 Overlaying radioactive blobs onto neutrino events

To simulate realistic detector conditions, a single Gaussian blob was added to each neutrino image. These images were then classified against blob-only background images using the Inception CNN.

The model was evaluated across a grid of blob widths σ and amplitudes A . The results are shown in Figure. 14, where accuracy, false positive rate, and false negative rate are presented across (σ, A) .

The classifier remains highly accurate across most of the parameter space, typically exceeding 0.98 and remaining close to unity for small and moderate blob sizes. The false negative rate remains below ~ 0.05 in most cases. A slight reduction in performance is

observed only for larger and brighter blobs (high σ and A), where accuracy decreases to ~ 0.90 .

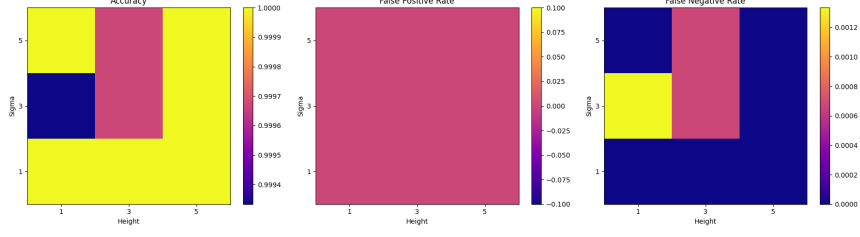


Figure 14. Classification performance of the Inception CNN for neutrino images with overlaid Gaussian blobs, tested against blob-only background images. Shown are the accuracy (left), false positive rate (centre), and false negative rate (right) as functions of blob width σ and height A .

5.2.3 Discussion

These results show that localised radioactive backgrounds do not significantly degrade classification performance. Unlike electronic noise, which affects the entire image, Gaussian blobs modify only a small region. The underlying neutrino structure therefore remains visible, allowing the classifier to distinguish signal from background using spatial features. This contrasts with the electronic noise results in Section 4, where performance degraded significantly at high noise levels.

6 Conclusion

This project investigated the classification of supernova neutrino interactions in liquid argon detectors using image-based machine learning models. All classifiers achieved near-perfect performance at low noise, but degraded as noise increased. Logistic regression and dense neural networks showed a steady decline, while the CNN performed best at low noise, reaching an accuracy of 0.962 at $\sigma = 5$, before collapsing to random guessing beyond $\sigma = 10$.

Alternative CNN architectures provided limited improvement. The Inception CNN showed some robustness at intermediate noise levels, achieving 0.751 accuracy at $\sigma = 10$, while the residual CNN showed no clear benefit. This indicates that architectural changes alone do not overcome the effects of noise.

Localised Gaussian blob backgrounds had minimal impact, with accuracies typically above 0.98. This reflects the fact that localised noise preserves the underlying signal structure, unlike electronic noise which affects the entire image.

Overall, performance is primarily limited by the signal-to-noise ratio. When noise becomes comparable to the signal, all models fail to extract reliable features. Future work should therefore focus on improving noise robustness rather than increasing model complexity.

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